

River Publishers Series in Digital Security and Forensics

APPLIED QUANTUM CRYPTANALYSIS

Alexei Petrenko



River Publishers

Applied Quantum Cryptanalysis

RIVER PUBLISHERS SERIES IN DIGITAL SECURITY AND FORENSICS

Series Editors

ANAND R. PRASAD

*Deloitte Tohmatu Cyber LLC,
Japan*

R. CHANDRAMOULI

*Stevens Institute of Technology,
USA*

ABDERRAHIM BENSLIMANE

*University of Avignon,
France*

The “River Publishers Series in Security and Digital Forensics” is a series of comprehensive academic and professional books which focus on the theory and applications of Cyber Security, including Data Security, Mobile and Network Security, Cryptography and Digital Forensics. Topics in Prevention and Threat Management are also included in the scope of the book series, as are general business Standards in this domain.

Books published in the series include research monographs, edited volumes, handbooks and textbooks. The books provide professionals, researchers, educators, and advanced students in the field with an invaluable insight into the latest research and developments.

Topics covered in the series include-

- Blockchain for secure Transactions
- Cryptography
- Cyber Security
- Data and App Security
- Digital Forensics
- Hardware Security
- IoT Security
- Mobile Security
- Network Security
- Privacy
- Software Security
- Standardization
- Threat Management

For a list of other books in this series, visit www.riverpublishers.com

Applied Quantum Cryptanalysis

Author

Alexei Petrenko

Innopolis University, Russia

Editor

Sergei Petrenko

Innopolis University, Russia



Published 2023 by River Publishers
River Publishers
Alsbjergvej 10, 9260 Gistrup, Denmark
www.riverpublishers.com

Distributed exclusively by Routledge
4 Park Square, Milton Park, Abingdon, Oxon OX14 4RN
605 Third Avenue, New York, NY 10017, USA

Applied Quantum Cryptanalysis / Alexei Petrenko and Sergei Petrenko.

©2023 River Publishers. All rights reserved. No part of this publication may be reproduced, stored in a retrieval systems, or transmitted in any form or by any means, mechanical, photocopying, recording or otherwise, without prior written permission of the publishers.

Routledge is an imprint of the Taylor & Francis Group, an informa business

ISBN 978-87-7022-793-3 (print)
ISBN 978-10-0087-946-9 (online)
ISBN 978-1-003-39287-3 (ebook master)

While every effort is made to provide dependable information, the publisher, authors, and editors cannot be held responsible for any errors or omissions.

Contents

Foreword	ix
Preface	xvii
Acknowledgments	xix
List of Figures	xxi
List of Tables	xxv
List of Abbreviations	xxvii
Glossary	xxxiii
Introduction	1
1 The Relevance of Quantum Cryptanalysis	7
1.1 National Quantum Programs	7
1.2 The Available Scientific Groundwork	18
1.3 Limiting Possibilities of Cryptanalysis Methods	36
1.4 The Quantum Threat of Blockchain	40
1.5 Requirements for Quantum Cryptanalysis	48
1.6 Verbal Statement of the Research Problem	61
Conclusions on Chapter 1	63
2 Implementation of the Shor Algorithm on a Quantum Circuit	67
2.1 Applicability of Fourier transforms	67
2.2 Reformulation of the Factorization Problem	72
2.3 Shor Algorithm Modification	75
2.4 Estimating the Quantum Resources Needed	79
2.5 Example Implementation of Shor's Algorithm	82
2.6 Features of the Implementation of the Shor Algorithm	87
2.7 The Mathematical Task Definition	93

Conclusions on Chapter 2	95
3 Development of Quantum Cryptanalysis Algorithms	97
3.1 Assessment of the Durability of Modern Cryptographic Algorithms to Clarify the Cryptanalysis Tasks of Asymmetric Encryption Schemes (RSA or El Gamal) and Digital Signature (DSA, ECDSA, or RSA-PSS) in a Quantum Computing Model	97
3.2 Implementation of Cryptanalysis Algorithms	100
3.3 Implementation of the Shor Factorization Algorithm	108
3.4 Implementation of the Grover's Search Algorithm	114
3.5 Development of an Algorithm for Recovering a Symmetric Encryption Key	117
3.6 Development of an Algorithm for Cryptanalysis of the RSA Asymmetric Encryption System.	121
3.7 Development of an Algorithm for Cryptanalysis of the El Gamal System	125
Conclusions on Chapter 3	128
4 Development of the "Q-Cryptanalysis" Platform	129
4.1 Features of the Implementation of the "Q-Cryptanalysis" Platform	129
4.1.1 Q-cryptanalysis platform development tools	129
4.1.2 Q-cryptanalysis platform architecture	131
4.1.3 Example of "Q-cryptanalysis" platform operation	131
4.2 Features of the used Quantum Computing Model.	132
4.3 Testing of the "Q-Cryptanalysis" Platform	143
4.4 The Evaluation of the Effectiveness of the "Q-Cryptanalysis" Platform.	145
4.5 Possible Development Directions of the "Q-Cryptanalysis" Platform.	148
4.6 A Feasibility Study for the Development of the "Q-Cryptanalysis" Platform.	149
Conclusions on Chapter 4	152
Conclusion	155
References	159

List of Open Training Courses in WWW	173
Websites of Manufacturers Of Quantum Computers And Simulators	175
Archives of Quantum Algorithms and Simulators	176
List of Well-known National Quantum Programs.	176
Index	179
About the Author	181
Publications	182
Projects	182



Taylor & Francis

Taylor & Francis Group

<http://taylorandfrancis.com>

Foreword

Dear readers!

The modern development level of information and communication technologies (ICT) realizes the opportunity to take industrial production and scientific research in information security to a fundamentally higher plane, but the effectiveness of such a transition directly depends on the availability of highly qualified specialists. About 5000 Russian information security specialists graduate every year, whereas the actual industrial demand is estimated at 21,000 per year until 2025. For this reason, the Russian Ministry of Education and Science, along with executive governmental bodies, has created a high-level training program, which they continually develop, for state information security employees. This initiative includes 170 universities, 40 institutions of continuing education, and 50 schools of secondary vocational training. In evaluating the universities' performance over 30 academic disciplines, information security has scored the highest for three consecutive years on the Russian Unified State Examination. In addition, employee training subsystems operating in the framework of the *Russian Federal Security Service*, the *Russian Ministry of Defense*, the *Russian Federal Protective Service*, *Russian Federal Service for Technical and Export Control*, and the *Russian Emergencies Ministry of Emergency Situations* are similar to the general system for training information security specialists at the *Russian Ministry of Education and Science*, which trains personnel according to the specific needs of individual departments.

Yet, there remains the well-known problem that the vast majority of educational programs in information security struggle to keep pace with the rapid development in the ICT sphere, where significant changes occur every 6 months. As a result, existing curricula and programs do not properly train graduates for the practical reality of what it means to efficiently solve modern information security problems. For this reason, graduates often find themselves lacking the actual skills in demand on the job market. In order to ensure that education in this field truly satisfies modern industrial demands, *Innopolis University* students and course participants complete

actual information security tasks for commercial companies as well as governmental bodies (e.g., for the universities *over 100 industrial partners*).

Also, *Innopolis University* students participate in domestic and international computer security competitions, e.g., the game *Capture the Flag (CTF)*, considered to be among the most authoritative in the world.

Currently, *Innopolis University* trains information security specialists in “*Computer Science and Engineering*” (MA program in *Secure Systems and Network Design*). The program is based on the *University of Amsterdam’s “System and Network Engineering” program with its focus on information security*. In 2013, it was ranked as the best MA program for IT in the *Netherlands (Keuzegids Masters 2013)*, and in 2015, it won the award for best educational program (*Keuzegids Masters 2015*). The *University of Amsterdam* is one of the partners of *Innopolis University* and is included in the top 50 universities of the world (*QS World university rankings, 2014/2015*).

An essential feature of this program is that *Innopolis University* students take part in relevant research and scientific-technical projects from the beginning of their studies. In solving computer security tasks, students have access to the scientific-technical potential of *3 institutes, 14 research laboratories, and 5 research centers* engaged in advanced IT research and development at *Innopolis University*. This partnership also extends to *Innopolis University’s* academic faculty, both pedagogic and research-oriented, which numbers more than *100 world-class specialists*. The information security education at *Innopolis University* meets the core curriculum requirements set out in the *State Educational Standards for Higher Professional Education 075 5000 “Information Security”* in the following degrees: “*Computer Security*,” “*Organization and Technology of Information Security*,” “*Complex Software Security*,” “*Complex Information Security of Automated Systems*,” and “*Information Security of Telecommunication Systems*.” At the same time, high priority is given to practical security issues of high industrial relevance; however, given the relative novelty of these needs, they remain insufficiently addressed in the curricula of most Russian universities and programs. These issues include the following:

- applied quantum cryptanalysis;
- quantum and post-quantum cryptography;
- quantum computers and computing;
- quantum communications;
- computer emergency response team (CERT) based on groundbreaking cognitive technologies;

- trusted cognitive supercomputer and ultra-high-performance technologies;
- adaptive security architecture technologies;
- intelligent technologies for ensuring information security based on Big Data and stream processing (*Big Data + ETL*);
- trusted device mesh technology and advanced system architecture;
- software-defined network (SDN) technology and network functions virtualization (NFV);
- hardware security module technology (HSM);
- trusted “cloud” and “foggy” computing, virtual domains;
- secure mobile technologies of 5G and 6G generations;
- organization and delivery of national and international cyber-training sessions;
- technologies for automated situation and opponent behavior modeling (*WarGaming*);
- technologies for dynamic analysis of program code and analytical verification, etc.

The current edition of the “**applied quantum cryptanalysis**” was written by *Alexei Petrenko, Ph.D., Associate Professor, Innopolis University*. The work of this author has significantly contributed to the creation of a national training system for highly qualified employees in the field of computer and data security technologies. This book sets out a notion of responsibility in training highly qualified specialists at the international level and in establishing a solid scientific foundation, which is prerequisite for any effective application of cybersecurity technologies.

Professor Alexander Tormasov
Rector Innopolis University,
Dr. Sc. (Phys. Math.)

Dear readers!

Today, we can witness an explosive growth of attention to Q-computing. Q-computing technologies, along with artificial intelligence (AI) and machine learning (ML) technologies, cloud and foggy computing, as well as technologies for collecting and streaming processing of Big Data and ETL, are constantly leading the lists of “end-to-end” information technologies for the digital economy of technologically developed countries of the world. One of the main reasons for this state of affairs is the potential ability of quantum computers to solve some computational problems more efficiently than any of the most modern classical computers of the von Neumann architecture (supercomputers). The most expressive and interesting, from an applied point of view, examples of such problems are integer factorization, effectively performed by Shor’s quantum algorithm, as well as record search in an unordered database, effectively solved by Grover’s algorithm.

In 1994, the mathematician Peter Shor, who worked at AT&T Bell Laboratories at that time, developed an algorithm solving the factorization problem in polynomial time (hence, the number of gates is polynomial) and on a polynomial number of qubits, while classical algorithms solve it in superpolynomial (subexponential) time. Therefore, as soon as a quantum computer with a sufficient number of qubits is created, all modern cryptography will be at risk of compromise. Actually, it will be compromised immediately, since any information, hidden using this approach, can be opened by any person who has access to such a quantum computer.

Shor’s algorithm differs from other well-known quantum algorithms in terms of having serious applied significance and is more complex in terms of mathematics and architecture. Two computational paradigms are involved in its implementation – the classical part prepares input data for the Shor algorithm and also manages cycles and returns in order to find the desired result; the quantum part executes a linear sequence of unitary transformations over specially prepared states of input qubits. The essence of the Shor factorization algorithm is to reduce the factorization problem to the problem of finding the period of the function. If the period of the function is known, then the mentioned factorization is carried out using the Euclid algorithm in polynomial time on a classical computer. The quantum part of the factorization algorithm is just looking for the period of the function. And the classical part of the algorithm first prepares the function in a special way and then checks the period found by the quantum part for sufficiency to solve the problem. If the period is found correctly (the algorithm is probabilistic; so it may not find what is required), then the problem is solved. If not, then the quantum part of the algorithm is to be run again. And, since checking the solution correctness

for the factorization problem is simple enough (multiplication of two numbers and comparison with the third), then this part of the algorithm can be ignored when calculating complexity.

It should be recognized that among the known quantum algorithms (more than 40), Shor's algorithm has generated a huge stir, and it can even be argued that because of the mentioned algorithm, a new computational model based on the laws of quantum mechanics has received such wide development. The thing is that numerous modern algorithms and cryptography systems are based on the hypothesis of algorithmic complexity of the number factorization problem. At the same time, scientists working in the cryptography believe that the National Security Agency (NSA)¹ and other intelligence agencies around the world have accumulated a huge amount of encrypted data from the Internet, which today cannot be decrypted by modern means. These data are stored and replenished, and it would be reasonable to assume that the NSA will be able to decrypt them when they get the appropriate quantum computer at their disposal. In such a scenario, not only the personal correspondence of citizens several decades ago will be at quantum threat, but the current correspondence, which we previously considered to be reliably protected, will be at risk. Gilles Brassard, a well-known computer science professor at the University of Montreal, is even more categorical² – “It would be absolute madness to believe that somewhere there is no one, and maybe a lot of those who record all network traffic and just wait for a technique that can crack all the old ciphers to appear. Therefore, although there is not yet a quantum computer sufficient for these purposes, and even if it is not developed within the next 5-10 years, as soon as it appears, all your correspondence that you sent from the first day using these classical encryption methods will be compromised, i.e., available to someone to whom it was not intended.”³

Thus, Shor's quantum algorithm solves factorization and discrete logarithm problems and can be used for cryptanalysis of most practically applicable cryptosystems (RSA, DSA, ECDSA, and GOST R 34.10). It is expected that in the next five years, quantum computers will surpass classical computers of the von Neumann architecture in solving the cryptanalysis problem, including cryptanalysis of the RSA cryptosystem (one of the most common systems of asymmetric encryption, named after its authors – Ron Rivest, Adi Shamir, and Leonard Adleman). By 2025, quantum computers will be able to

¹ <https://www.nsa.gov>

² <http://www.iro.umontreal.ca/~brassard/web/en/>

³ <https://spkurdyumov.ru/uploads/2016/04/kvantovyj-vzlom.pdf>

effectively crack RSA with a key length of 2048 bits (the minimum recommended by international cryptographic standards).

At the moment, using the Shor algorithm, the numbers $15 = 3 \times 5$ and $21 = 3 \times 7$ have been successfully factorized on quantum computers.⁴ Also, a 4-qubit adiabatic quantum computer was successfully adapted to solve the factorization problem, factoring the number $143 = 11 \times 13$ and the number $56,153 = 233 \times 241$.⁵ Curiously, the factorization of a larger number at first went unnoticed by researchers, and only two years later was it shown that a whole class of numbers was factorized during the experiment. Next, the number 200,099 was factorized by quantum annealing on a D-Wave 2X computer.⁶ The next interesting result was the factorization of the number 291,311⁷ using a quantum computer based on the principles of nuclear magnetic resonance. And the record factorized number at the current time is the number $1,099,551,473,989 = 1,048,589 \times 1,048,601$.⁸

Note that the practical results of applying Grover and Simon algorithms to the analysis of model cryptosystems are still poorly understood, since the implementation of such systems requires a large number of quantum gates, which is not available at the current level of development of Q-technology. However, selective examples of the implementation of the Grover method on model problems and its implementation, including on the IBM cloud quantum computer, are presented in many works.^{9, 10} The implementation of Simon's algorithm is described in the technical guidance on quantum algorithms.¹¹ It is also interesting that the most impressive results on factorization were obtained using quantum computers implementing models that were previously considered not quite suitable cryptanalysis problems, such as, for example, the quantum annealing model.

Thus, the development of the algorithmic base is able to shorten the time of the appearance of efficient quantum computers relative to our estimate. Moreover, the prospects for the appearance of a "practical" quantum

⁴ <https://research-information.bris.ac.uk/en/publications/experimental-realization-of-shors-quantum-factoring-algorithm-usi>

⁵ <https://arxiv.org/abs/1411.6758>

⁶ <https://arxiv.org/abs/1604.05796>

⁷ <https://www.researchgate.net/scientific-contributions/Richard-Tanburn-2079794789>

⁸ <https://4627su41pzrvhaad34118k3y-wpengine.netdna-ssl.com/wp-content/uploads/2020/12/Analyzing-the-Performance-of-Variational-Quantum-Factoring-on-a-Superconducting-Quantum-Processor.pdf>

⁹ <https://cis.temple.edu/~boji/papers/REU2018.pdf>

¹⁰ <https://arxiv.org/pdf/1804.03719.pdf>

¹¹ <https://qiskit.org/textbook/ch-algorithms/simon.html>

computer capable of performing cryptanalysis tasks become even closer, if we take into account the results of IBM's development of quantum processors. So, in November 2021, IBM introduced the 127-qubit Eagle processor, and by 2023 predicts overcoming the 1000-qubit limit.¹² At one time, Google researchers showed that about 20,000,000 physical (available at the current level of technology) qubits are enough for effective cryptanalysis of RSA.¹³ Taking into account the possibility of efficient parallelization of calculations between several devices with a significantly smaller number of qubits, demonstrated in this work, IBM's achievements convincingly demonstrate the realism of the quantum threat.

The foregoing raises the urgency of the presented “**Applied Quantum Cryptanalysis**” monograph. I hope that this book will be a very valuable tool for the development and formation of highly qualified specialists of a new class in the field of information technology and cybersecurity.

Professor Sergei Petrenko

Head of the Information Security Center at Innopolis University
Dr. Sc. (Tech.)

¹² <https://newsroom.ibm.com/2021-11-16-IBM-Unveils-Breakthrough-127-Qubit-Quantum-Processor>

¹³ <https://arxiv.org/pdf/1905.09749.pdf>



Taylor & Francis

Taylor & Francis Group

<http://taylorandfrancis.com>

Preface

“Quantum computers will make modern cryptography methods hopelessly obsolete. And then what will happen?”

American science and nature writer. Editor at Discover Magazine.
“Series editor” of The Best American Science and Nature Writing
yearly anthology since 2002.

Tim Folger

This monograph contains the best practice of solving problems of quantum cryptanalysis to improve cybersecurity and resilience of the digital economy. The book discusses the well-known and author’s software implementations of promising quantum Shore algorithms, Grover, Simon, and others. Shor’s algorithm provides exponential acceleration of solving factorization problems, discrete logarithm (DLP), and elliptic curve discrete logarithm problems (ECDLP). The mentioned tasks are widely used in TLS, SSH, or IPsec cryptographic applications of Internet/Intranet and IIoT/IoT networks, communication protocols based on Diffie–Hellman key agreements (depend on the strength of DLP or ECDLP), digital signature algorithms (DSA, ECDSA, and RSA-PSS), public key encryption algorithms (El Gamal and RSA-OAEP), etc. In other words, Shor’s quantum algorithm is potentially capable of violating these algorithms, and with them, all the mechanisms of public key cryptography deployed in cyberspace.

The book will be useful for chief information officers (CIO) and chief information security officers (CISO), certified information systems auditors (CISA), managers of the top echelon of companies responsible for ensuring security and cyber resilience of business, as well as teachers and students of MBA, CIO, and CSO programs, students, and postgraduates of relevant specialties.



Taylor & Francis

Taylor & Francis Group

<http://taylorandfrancis.com>

Acknowledgments

The author would like to thank Professor Alexander G. Tormasov (Rector Innopolis University) and Professor Sergei A. Petrenko (Head of the Information Security Center at Innopolis University) for the foreword and support.

The author would like to sincerely thank Professor Nikolai A. Moldovyan (Russian Academy of Sciences, RAS), Professor Alexander G. Lomako, and Professor Igor A. Sheremet (Russian Foundation for Basic Research, RFBR) for their valuable advice and comments on the manuscript, the editing of which contributed to the improvement of its quality.

The author would like to thank Professor Alexander G. Lomako and Professor Alexey S. Markov (Bauman Moscow State Technical University) for the positive review and semantic editing of the monograph.

The author would like to thank his friends and colleagues: Alexander Romanchenko, Sergei Grebnev, and Aleksey Fedorov (qapp. tech), Vladimir Belsky and Ivan Chizhov (Kryptonite.ru), and Dmitry Khovratovich and Sergei Tikhomirov (Chaincode Labs.) for their support and attention to the work.

The author would like to express special gratitude to Nikolai Nikiforov – Minister of Informatization and Communication of Russian Federation, Roman Shayhutdinov – Deputy Prime Minister of the Republic of Tatarstan, Minister of Informatization and Communication of the Republic of Tatarstan, and Prof. Dr. Alexander S. Holevo – Academician of the Russian Academy of Sciences (RAS).

The author would also like to thank Elvira Khismatullina for translating the original text into English as well as Rajeev Prasad – publisher at River Publishers for providing the opportunity to publish the book and Junko Nagajima – production coordinator who tirelessly worked through several iterations of corrections for assembling the diverse contributions into a homogeneous final version. The reported study was funded by RFBR, project number 20-04-60080 “*Models and methods for ensuring the sustainability of*

xx *Acknowledgments*

society's social and technical systems in the face of viral epidemics such as the COVID-19 pandemic based on acquired immunity.”

Alexei Petrenko
Associate Professor Innopolis University, Ph.D.
fataw11b@gmail.com

List of Figures

Figure 1.1	The main directions of quantum computing development.	9
Figure 1.2	Prerequisites of the US National Quantum Initiative (2018).. . . .	11
Figure 1.3	Quantum Satellite Communications of China (Micius, 2016).	12
Figure 1.4	The role and place of DARPA and IARPA in the implementation of the US National Quantum Initiative (2018).	15
Figure 1.5	Budget line for the US National Quantum Initiative..	16
Figure 1.6	The priority of the development of quantum algorithms in the interests of the US Department of Defense.. . . .	17
Figure 1.7	Estimation of the complexity of decomposing a large integer into prime factors.. . . .	22
Figure 1.8	The relevance of the software-technical implementation of the Shor quantum algorithm for opening asymmetric encryption schemes.	24
Figure 1.9	Extrapolation of RSA number factorization capabilities.	25
Figure 1.10	The first pilot solutions of post-quantum cryptography.	31
Figure 1.11	Evaluation of the post-quantum cryptography market.	31
Figure 1.12	Possible solutions to post-quantum cryptography, Qapp.tech.. . . .	32
Figure 1.13	Possible options for integrating the PQLR SDK, Qapp.tech.. . . .	32
Figure 1.14	How Qtunnel and Qapp.tech work.	33
Figure 1.15	Classification of known cryptanalysis methods. . . .	33
Figure 1.16	The role and place of quantum cryptanalysis methods.. . . .	37

Figure 1.17	Risks of a quantum threat to blockchain systems. . .	41
Figure 1.18	Data with a long life-cycle is more vulnerable (data interception now, delayed data hacking). . . .	41
Figure 1.19	A typical scheme of the functioning of the blockchain.	43
Figure 1.20	The diagram of Bitcoin blockchain.	46
Figure 1.21	Existing ways to connect a personal computer to demonstration prototypes of real quantum computers.	51
Figure 1.22	The area of guaranteed solution of the quantum cryptanalysis problem with the required performance.	63
Figure 2.1	Calculation scheme of the FFT transformation of size 4.. . . .	69
Figure 2.2	An example of a quantum circuit on three qubits. . .	84
Figure 2.3	Four-qubit quantum Fourier transformation.	85
Figure 2.4	Function values.. . . .	86
Figure 2.5	Quantum scheme for calculating the function.	88
Figure 2.6	Quantum scheme for the algorithm of finding the order.	89
Figure 2.7	Probability distribution of getting different y values..	91
Figure 3.1	Shor's algorithm in the gate representation, 1994. . .	99
Figure 3.2	Grover's algorithm in the gate representation, 1996. .	100
Figure 3.3	States of a qubit in the form of a Bloch sphere. . . .	102
Figure 3.4	Example of basic one-bit gates.. . . .	103
Figure 3.5	Application of the CNOT gate (operator).	104
Figure 3.6	Diagram of the Shor factorization algorithm.. . . .	111
Figure 3.7	Diagram of Grover's quantum search algorithm. . . .	116
Figure 3.8	Quantum algorithm for symmetric encryption key recovery based on the message text and ciphertext has been developed.	119
Figure 3.9	Quantum algorithm for cryptanalysis of the RSA asymmetric encryption system.	123
Figure 3.10	Quantum cryptanalysis algorithm of El Gamal system.	126
Figure 4.1	Structural and functional diagram of "Q-cryptanalysis" platforms.	132
Figure 4.2	"Q-cryptanalysis" flowchart.	133
Figure 4.3	"Q-cryptanalysis" output window when entering a factorizable number.	134
Figure 4.4	Auxiliary number request window.	134

Figure 4.5	Request window for the number operation calculations.	135
Figure 4.6	Example of the result of the software cryptanalysis complex “Q-cryptanalysis.”.	136
Figure 4.7	A typical scheme of asymmetric encryption.	137
Figure 4.8	Explanations for assessing the durability of the studied cryptosystems.	137
Figure 4.9	The main stages of the proposed quantum Shor factorization algorithm.	139
Figure 4.10	Finding the required number of qubits, as a criterion for the success of solving the cryptanalysis problem.	140
Figure 4.11	An example of solving the factorization problem by quantum computing.	141
Figure 4.12	Gate representation of the quantum Fourier transformation.	142
Figure 4.13	Quantum representation of the modified quantum Shor factorization algorithm.	142
Figure 4.14	Example of quantum gate modeling.	142
Figure 4.15	Testing of the developed “Q-cryptanalysis” platform in the <i>IBM Q System One</i> environment (16-qubit) and a fifth generation supercomputer.	143
Figure 4.16	The total gain from the proposed quantum implementations of the Shor algorithm in the IBM Q hybrid environment and a fifth generation supercomputer.	146
Figure 4.17	Representations of Shor’s quantum algorithm for IBM Q hybrid environment and fifth generation supercomputers.	146
Figure 4.18	Comparative time analysis of cryptanalysis of schemes of asymmetric encryption (<i>RSA</i> or <i>El Gamal</i>) and digital signature (<i>DSA</i> , <i>ECDSA</i> , or <i>RSA-PSS</i>).	147
Figure 4.19	Increasing the share of quantum cryptanalysis algorithms of asymmetric encryption (<i>RSA</i> or <i>El Gamal</i>) and digital signature (<i>DSA</i> , <i>ECDSA</i> , or <i>RSA-PSS</i>).	147
Figure 4.20	Promising platform architecture of the “Q-cryptanalysis.”.	148



Taylor & Francis

Taylor & Francis Group

<http://taylorandfrancis.com>

List of Tables

Table 1.1	List of well-known national quantum programs. . . .	8
Table 1.2	The main stages of the formation of a scientific foundation for solving practical problems of cryptography and cryptanalysis.	19
Table 1.3	Basic quantum algorithms for solving cryptanalysis problems.	23
Table 1.4	The main current results of factorization of numbers used as a public key in the RSA public key cryptosystem.	24
Table 1.5	The most common models of quantum computers, 2021.	27
Table 1.6	Estimates of the cryptographic strength of the most common crypto algorithms in the USA and EU countries.	28
Table 1.7	List of potentially vulnerable blockchain systems. . .	28
Table 1.8	PQLR SDK – a library of quantum-stable algorithms integrated with OpenSSL and Qapp.tech.. . . .	34
Table 1.9	Evaluation of cryptographic strength of encryption algorithms.	38
Table 1.10	Known integer factorization methods.	49
Table 1.11	Results of comparison of mathematical software packages for modeling quantum algorithms on quantum circuits.	49
Table 1.12	Results of comparison of quantum computing emulators.. . . .	50
Table 1.13	Evaluation of the accuracy of quantum operations.. .	58
Table 1.14	Results of comparison of platforms for connection to quantum computers IBM Q (up to 20 qubits), Rigetti (up to 8 qubits), and ETH Zurich (up to 16 qubits). .	60
Table 1.15	Asymmetric encryption and digital signature schemes for opening by quantum cryptanalysis algorithms. . .	62

Table 2.1	Resources for quantum problem solving – factorization and discrete logarithm problems.	81
Table 2.2	Resources for quantum problem solution finding the key of a symmetric cryptosystem.	82
Table 2.3	Characteristics of cryptosystem durability.	94
Table 3.1	Matching the length of symmetric keys and asymmetric ciphers with the same cryptographic strength.	100
Table 3.2	Comparison of the complexity of the main tasks of cryptanalysis in classical and quantum models of computing.	101
Table 3.3	Time gain for solving problems of quantum circuits on quantum computers.. . . .	101
Table 3.4	Basic elementary transformations (or quantum gates).	107
Table 4.1	Time complexity of algorithms, solving factorization, and discrete logarithm problems.	138
Table 4.2	Testing stages of the “Q-cryptanalysis” platform. . .	145
Table 4.3	Remuneration of platform developers.	150
Table 4.4	Material costs for the development of the complex. .	151

List of Abbreviations

AES	Advanced Encryption Standard (as specified in FIPS 197).
AES-CMAC	The AES cipher-based MAC mode (as specified in SP 800-38B).
ASN.1	Abstract Syntax Notation One.
$C(i_e)$	Notation for a category of key-establishment schemes in which i ephemeral key pairs are used, where $i \in \{0, 1, 2\}$.
$C(i_e, j_s)$	Notation for a subcategory of key-establishment schemes in which i ephemeral key pairs and j static key pairs are used. In this recommendation, schemes in the subcategories $C(0_e, 2_s)$, $C(1_e, 2_s)$, $C(1_e, 1_s)$, $C(2_e, 0_s)$, and $C(2_e, 2_s)$ are defined.
CA	Certification authority.
CDH	The cofactor <i>ECC Diffie–Hellman</i> key-agreement primitive.
CSP	Critical security parameter.
DH	The (non-cofactor) <i>FFC Diffie–Hellman</i> key-agreement primitive.
DLC	Discrete logarithm cryptography, which is composed of both <i>finite field cryptography (FFC)</i> and <i>elliptic curve cryptography (ECC)</i> .
EC	Elliptic curve.
ECC	Elliptic curve cryptography; the public key cryptographic methods using operations in an elliptic curve group.
FF	Finite field.
FFC	Finite field cryptography; the public key cryptographic methods using operations in a multiplicative group of a finite field.
ID	The bit string denoting the identifier associated with an entity.
KC	Key confirmation.
KDM	Key-derivation method.

KM	Keying material.
KMAC	The <i>KECCAK</i> -based MAC (as specified in SP 800-185).
len(x)	The bit length of the shortest base-two representation of the positive integer x , i.e., $\text{len}(x) = \lfloor \log_2(x) \rfloor + 1$.
MAC	Message authentication code.
MAC (<i>MacKey</i>, <i>MacData</i>)	A MAC algorithm with <i>MacKey</i> as the key, and <i>MacData</i> as the data.
MacOutputBits	The length of the MAC output block in bits.
MacTag	An MAC tag.
MacTagBits	The length of the MacTag in bits.
MQV	The Menezes–Qu–Vanstone key-agreement primitive.
Null	The empty bit string.
RBG	Random bit generator.
SHA	Secure hash algorithm (as specified in FIPS 180 and FIPS 202).
$T_{\text{bitLen}}(X)$	A truncation function that outputs the most significant (i.e., leftmost) bitLen bits of the input bit string, X , when the bit length of X is greater than bitLen ; otherwise, the function outputs X . For example, $T_2(1011) = 10$, $T_3(1011) = 101$, $T_4(1011) = 1011$, and $T_5(1011) = 1011$.
TTP	Trusted third party.
U, V	Represents the two parties in a (pair-wise) key-establishment scheme.
{ }	In this recommendation, the curly braces { } are used in the following three situations: (1) { x } is used to indicate that the inclusion of x is optional; for example, the notation “Input: w { x }, y , and z ” implies that the inclusion of x as an input is optional. (2) If both X and Y are binary strings, the notation of binary string “ $Y\{\ X\}$ ” implies that the concatenation of string X is optional. (3) { $x_1, x_2, \dots, x_k$ } indicates a set with elements $x_1, x_2, \dots, x_k$.
$X \ Y$	The concatenation of two-bit strings X and Y . For example, $11001 \ 010 = 11001010$.
[a, b]	The set of integers x , such that $a \leq x \leq b$.
$\lceil x \rceil$	The ceiling of x ; the smallest integer $\geq x$. For example, $\lceil 5 \rceil = 5$, $\lceil 5.3 \rceil = 6$.
$\lfloor x \rfloor$	The floor of x ; the greatest integer that does not exceed x . For example, $\lfloor 2.1 \rfloor = 2$, and $\lfloor 4 \rfloor = 4$.

Z	A shared secret (represented as a byte string) that is used to derive secret keying material using a key-derivation method.
Z_e	A component of the shared secret (represented as a byte string) that is computed using ephemeral keys in a Diffie–Hellman primitive.
Z_s	A component of the shared secret (represented as a byte string) that is computed using static keys in a Diffie–Hellman primitive.
$GF(p)$	The finite field with p elements, where p is an (odd) prime number. The elements of $GF(p)$ can be represented by the set of integers $\{0, 1, \dots, p - 1\}$. The addition and multiplication operations for $GF(p)$ can be realized by performing the corresponding integer operations and reducing the results modulo p .
$GF(p)^*$	The multiplicative group of non-zero field elements in $GF(p)$.
g	An FFC domain parameter; the selected generator of the multiplicative subgroup of prime order q in $GF(p)^*$.
$k \bmod p$	The modular reduction of the (arbitrary) integer k by the (positive) integer p (the modulus). For the purposes of this recommendation, $j = k \bmod p$ is the unique integer satisfying the following two conditions: $0 \leq j < p$, and $k - j$ is a multiple of p . In short, $j = k - \lfloor k/p \rfloor p$.
p	An FFC domain parameter; an odd prime number that determines the size of the finite field $GF(p)$.
<i>counter</i>	An optional FFC domain parameter; a value that may be output during domain parameter generation to provide assurance at a later time that the resulting domain parameters were generated using a canonical process.
q	When used as an FFC domain parameter, q is the (odd) prime number equal to the order of the multiplicative subgroup of $GF(p)^*$ generated by g . Note that q is a divisor of $p - 1$.
r_U, r_V	The ephemeral private keys of party U and party V , respectively. These are integers in the interval $[1, q - 1]$. (In some instances, r_U and/or r_V may be restricted to a subinterval of the form $[1, 2N - 1]$).
t_U, t_V	The ephemeral public keys of party U and party V , respectively. These are integers in the interval $[2, p - 2]$.

SEED	An <i>FFC</i> domain parameter; an initialization value that is used during domain parameter generation that can also be used later to provide assurance that the resulting domain parameters were generated using an approved process.
x_U, x_V	The static private keys of party <i>U</i> and party <i>V</i> , respectively. These are integers in the interval $[1, q - 1]$ (in some instances, x_U and/or x_V may be restricted to a subinterval of the form $[1, 2^N - 1]$).
y_U, y_V	The static public keys of party <i>U</i> and party <i>V</i> , respectively. These are integers in the interval $[2, p - 2]$.
a, b	ECC domain parameters; two elements in the finite field $GF(q)$ that define the (<i>Weierstrass</i>) equation of an elliptic curve, $y^2 = x^3 + ax + b$ when q is an odd prime p or $y^2 + xy = x^3 + ax^2 + b$ when $q = 2^m$ for some prime integer m .
$avf(Q)$	The associate value of the elliptic curve point Q .
$d_{e,U}, d_{e,V}$	The ephemeral private keys of party <i>U</i> and party <i>V</i> , respectively. These are integers in the interval $[1, n - 1]$.
$d_{s,U}, d_{s,V}$	The static private keys of party <i>U</i> and party <i>V</i> , respectively. These are integers in the interval $[1, n - 1]$.
FR	Field representation indicator (an <i>ECC</i> domain parameter); an indication of the basis used for representing field elements. If a polynomial basis representation is used for a field of order 2^m , then <i>FR</i> indicates the reduction polynomial (a trinomial or a pentanomial); otherwise, <i>FR</i> is null.
G	An <i>ECC</i> domain parameter, which is a distinguished (affine) point in an elliptic curve group that generates a subgroup of prime order n .
GF(q)	The finite field with q elements, where either q is an odd prime p , or q is equal to 2^m for some prime integer m . The elements of $GF(q)$ are represented by the set of integers $\{0, 1, \dots, p - 1\}$ in the case that q is an odd prime p , or as bit strings of length m bits in the case that $q = 2^m$.
h	An <i>ECC</i> domain parameter; the cofactor, a positive integer that is equal to the order of the elliptic curve group, divided by the order of the cyclic subgroup generated by the distinguished point G . That is, nh is the order of the elliptic curve, where n is the order of the cyclic subgroup generated by the distinguished point G .

n	An <i>ECC</i> domain parameter; a prime that is the order of the cyclic subgroup generated by the distinguished point G .
$\emptyset$	The (additive) identity element of an elliptic curve group; also called the “neutral point” of that group. $\emptyset$ is the unique element satisfying $Q + \emptyset = \emptyset + Q = Q$ for each Q in the group. For the (<i>Weierstrass</i>) elliptic curve groups considered in this recommendation, a special “ <i>point at infinity</i> ” serves as $\emptyset$.
q	When used as an <i>ECC</i> domain parameter, q is the field size. It is either an odd prime p or equal to 2^m for some prime integer m .
$\mathcal{Q}_{e,U}, \mathcal{Q}_{e,V}$	The ephemeral public keys of party U and party V , respectively. These are points on the elliptic curve that is defined by the domain parameters.
$\mathcal{Q}_{s,U}, \mathcal{Q}_{s,V}$	The static public keys of party U and party V , respectively. These are points on the elliptic curve that is defined by the domain parameters.
SEED	An optional <i>ECC</i> domain parameter; an initialization value that is used during domain parameter generation that can also be used later to provide assurance that the resulting domain parameters were generated using an approved process.
x_P, y_P	Elements of the finite field $GF(q)$ representing the x and y coordinates, respectively, of a point P .



Taylor & Francis

Taylor & Francis Group

<http://taylorandfrancis.com>

Glossary

Common Terms and Definitions

Active entity: A user or a process acting on behalf of a user. It is also referred to as a subject.

Algorithm: A clearly specified mathematical process for computation; a set of rules that, if followed, will give a prescribed result.

Approved: FIPS-approved or NIST-recommended. An algorithm or technique that is either (1) specified in a FIPS or NIST recommendation, or (2) adopted in a FIPS or NIST recommendation and specified either (a) in an appendix to the FIPS or NIST recommendation, or (b) in a document referenced by the FIPS or NIST recommendation.

Assumption: Used to indicate the conditions that are required to be true when an approved key-establishment scheme is executed in accordance with this recommendation.

Assurance of private key possession: Confidence that an entity possesses a private key corresponding to a public key.

Assurance of validity: Confidence that either a key or a set of domain parameters is arithmetically correct.

Big-endian: The property of a byte string having its bytes positioned in the order of decreasing significance. In particular, the leftmost (first) byte is the most significant byte (containing the most significant eight bits of the corresponding bit string) and the rightmost (last) byte is the least significant byte (containing the least significant eight bits of the corresponding bit string). For the purposes of this recommendation, it is assumed that the bits within each byte of a big-endian byte string are also positioned in the order of

decreasing significance (beginning with the most significant bit in the leftmost position and ending with the least significant bit in the rightmost position).

Binding: Assurance of the integrity of an asserted relationship between items of information that is provided by cryptographic means. Also see “Trusted association.”

Bit length: The length in bits of a bit string.

Bit string: An ordered sequence of 0s and 1s. Also known as a binary string.

Byte: A bit string consisting of eight bits.

Byte string: An ordered sequence of bytes.

Certificate authority (CA): The entity in a public key infrastructure (PKI) that is responsible for issuing public key certificates and exacting compliance to a PKI policy. Also known as a certification authority.

Cofactor: The order of the elliptic curve group divided by the (prime) order of the generator point (i.e., the base point) specified in the domain parameters.

Confidentiality: The property that sensitive information is not disclosed to unauthorized entities.

Critical security parameter (CSP): Security-related information whose disclosure or modification can compromise the security of a cryptographic module. Domain parameters, secret or private keys, shared secrets, key-derivation keys, intermediate values, and secret salts are examples of quantities that may be considered CSPs in this recommendation. See FIPS 140.

Cryptographic key (key): A parameter used with a cryptographic algorithm that determines its operation.

Cryptographic module: The set of hardware, software, and/or firmware that implements approved security functions (including cryptographic algorithms and key generation). See FIPS 140.

Destroy: In this recommendation, an action applied to a key or a piece of secret data. After a key or a piece of secret data is destroyed, no information about its value can be recovered. Also known as zeroization in FIPS 140.

Domain parameters: The parameters used with a cryptographic algorithm that are common to a domain of users.

Entity: An individual (person), organization, device, or process. “Party” is a synonym.

Ephemeral key pair: A key pair, consisting of a public key (i.e., an ephemeral public key) and a private key (i.e., an ephemeral private key) that is intended for a very short period of use. The key pair is ordinarily used in exactly one transaction of a cryptographic scheme. Contrast with a static key pair.

Fresh: Newly established keying material that is statistically independent of any previously established keying material.

Hash function: A function that maps a bit string of arbitrary length to a fixed-length bit string. Approved hash functions are expected to satisfy the following properties: (1) one-way: it is computationally infeasible to find any input that maps to any pre-specified output; (2) collision resistant: it is computationally infeasible to find any two distinct inputs that map to the same output.

Identifier: A bit string that is associated with a person, device, or organization. It may be an identifying name or may be something more abstract (for example, a string consisting of an IP address).

Integrity: A property whereby data has not been altered in an unauthorized manner since it was created, transmitted, or stored.

Key agreement: A (pair-wise) key-establishment procedure in which the resultant secret keying material is a function of information contributed by both participants so that neither party can predetermine the value of the secret keying material independently from the contributions of the other party. Contrast with key-transport.

Key-agreement transaction: An execution of a key-agreement scheme.

Key confirmation: A procedure to provide assurance to one party (the key-confirmation recipient) that another party (the key-confirmation provider) possesses the correct secret keying material and/or the shared secret from which that keying material is derived.

Key-confirmation provider: The party that provides assurance to the other party (the recipient) that the two parties have indeed established a shared secret or shared keying material.

Key-derivation function: As used in this recommendation, a function used to derive secret keying material from a shared secret and other information.

Key-derivation method: As used in this recommendation, a method used to derive secret keying material from a shared secret and other information. A key-derivation method may use a key-derivation function or a key-derivation procedure.

Key-derivation procedure: As used in this recommendation, a multi-step process that is used to derive keying material from a shared secret and other information.

Key establishment: The procedure that results in secret keying material that is shared among different parties.

Key-establishment key pair: A private/public key pair that is used in a key-establishment scheme. It can be a static key pair or an ephemeral key pair.

Key-establishment transaction: An instance of establishing secret keying material using a key-agreement or key-transport transaction.

Key-transport: A (pair-wise) key-establishment procedure whereby one party (the sender) selects a value for the secret keying material and then securely distributes that value to another party (the receiver). Contrast with key agreement.

Key-transport transaction: An execution of a key-transport scheme.

Key-wrapping: A method of protecting keying material (along with associated integrity information) that provides both confidentiality and integrity protection by using symmetric-key algorithms.

Key-wrapping key: In this recommendation, a key-wrapping key is a symmetric key established during a key-agreement transaction and used with a key-wrapping algorithm to protect the keying material to be transported.

Key-wrapping algorithm: An algorithm for protecting keying material that provides both confidentiality and integrity protection using a symmetric key-wrapping key.

Key-wrapping key: A symmetric key used with a key-wrapping algorithm to protect the keying material to be transported.

Keying material: Data that is represented as a binary string such that any non-overlapping segments of the string with the required lengths can be used as secret keys, secret initialization vectors, and other secret parameters.

MAC tag: Data obtained from the output of an MAC algorithm (possibly by truncation) that can be used by an entity to verify the integrity and the origination of the information used as input to the MAC algorithm.

Message authentication code (MAC) algorithm: A family of cryptographic functions that are parameterized by a symmetric key. Each of the functions can act on input data (called a “message”) of variable length to produce an output value of a specified length. The output value is called the MAC of the input message. An approved MAC algorithm is expected to satisfy the following property (for each of its supported security levels): it must be computationally infeasible to determine the (as yet unseen) MAC of a message without knowledge of the key, even if one has already seen the results of using that key to compute the MACs of other (different) messages. An MAC algorithm can be used to provide data-origin authentication and data-integrity protection. In this recommendation, an MAC algorithm is used for key confirmation; the use of MAC algorithms for key derivation is addressed in SP 800-56C.

Nonce: A time-varying value that has at most an acceptably small chance of repeating. For example, the nonce may be a random value that is generated anew for each use, a timestamp, a sequence number, or some combination of these.

Owner: For a static public key, static private key, and/ or the static key pair containing those components, the owner is the entity that is authorized to use the static private key corresponding to the static public key, whether that entity generated the static key pair itself or a trusted party generated the key pair for the entity. For an ephemeral public key, ephemeral private key, and/or ephemeral public key pair, the owner is the entity that generated the

ephemeral key pair and is authorized to use the ephemeral private key of the key pair.

Party: See “Entity.”

Prime number: An integer that is greater than 1 and divisible only by 1 and itself.

Primitive: A low-level cryptographic algorithm that is used as a basic building block for higher-level cryptographic operations or schemes.

Private key: A cryptographic key that is kept secret and is used with a public key cryptographic algorithm. A private key is associated with a public key.

Protocol: A set of rules used by two or more communicating entities that describe the message order and data structures for information exchanged between the entities.

Provider: A party that provides (1) a public key (e.g., in a certificate); (2) assurance, such as an assurance of the validity of a candidate public key or assurance of possession of the private key associated with a public key; or (3) key confirmation. Contrast with recipient.

Public key: A cryptographic key that may be made public and is used with a public key cryptographic algorithm. A public key is associated with a private key.

Public key certificate: A data structure that contains an entity’s identifier(s), the entity’s public key (including an indication of the associated set of domain parameters) and possibly other information, along with a signature on that data set that is generated by a trusted party, i.e., a certificate authority, thereby binding the public key to the included identifier(s).

Public key validation: The procedure whereby the recipient of a public key checks that the key conforms to the arithmetic requirements for such a key in order to thwart certain types of attacks.

Random nonce: A nonce containing a random-value component that is generated anew for each nonce.

Receiver: The party that receives secret keying material via a key-transport transaction. Contrast with sender.

Recipient: A party that (1) receives a public key; or (2) obtains assurance from an assurance provider (e.g., assurance of the validity of a candidate public key or assurance of possession of the private key corresponding to a public key); or (3) receives key confirmation from a key-confirmation provider.

Scheme: A set of unambiguously specified transformations that provide a (cryptographic) service when properly implemented and maintained. A scheme is a higher-level construct than a primitive and a lower-level construct than a protocol.

Security properties: The security features (e.g., replay protection, or key confirmation) that a cryptographic scheme may, or may not, provide.

Security strength (also, “bits of security”): A number associated with the amount of work (that is, the number of operations) that is required to break a cryptographic algorithm or system.

Sender: The party that sends secret keying material to the receiver in a key-transport transaction. Contrast with receiver.

Shall: This term is used to indicate a requirement that needs to be fulfilled to claim conformance to this recommendation. Note that shall may be coupled with not to become shall not.

Shared secret: A secret value that has been computed during a key-establishment scheme, which is known by both participants and is used as input to a key-derivation method to produce keying material.

Should: This term is used to indicate an important recommendation. Ignoring the recommendation could result in undesirable results. Note that should may be coupled with not to become should not.

Static key pair: A key pair, consisting of a private key (i.e., a static private key) and a public key (i.e., a static public key) that is intended for use for a relatively long period of time and is typically intended for use in multiple key-establishment transactions. Contrast with an ephemeral key pair.

Store and forward: A telecommunications technique in which information is sent to an intermediate station where it is kept and later sent to the final destination or to another intermediate station.

Support (a security strength): A security strength of s bits is said to be supported by a particular choice of algorithm, primitive, auxiliary function, parameters, etc., for use in the implementation of a cryptographic mechanism if that choice will not prevent the resulting implementation from attaining a security strength of at least s bits. In this recommendation, it is assumed that implementation choices are intended to support a security strength of 112 bits or more (see SP 800-57 and SP 800-131A).

Symmetric key: A cryptographic key that is shared between two or more entities and used with a cryptographic application to process information.

Symmetric-key algorithm: A cryptographic algorithm that uses secret keying material that is shared between authorized parties.

Targeted security strength: The security strength that is intended to be supported by one or more implementation-related choices (such as algorithms, primitives, auxiliary functions, parameter sizes, and/or actual parameters) for the purpose of instantiating a cryptographic mechanism. In this recommendation, it is assumed that the targeted security strength of any instantiation of an approved key-establishment scheme has a value greater than or equal to 112 bits and less than or equal to 256 bits.

Trusted association: Assurance of the integrity of an asserted relationship between items of information that may be provided by cryptographic or non-cryptographic (e.g., physical) means. Also see “Binding.”

Trusted party: A party that is trusted by an entity to faithfully perform certain services for that entity. An entity could be a trusted party for itself.

Trusted third party: A third party, such as a CA, that is trusted by its clients to perform certain services. By contrast, the two participants in a key-establishment transaction are considered to be the first and second parties.

Introduction

Currently, the national quantum programs supporting exploratory research in the field of quantum technologies are being implemented in 17 technologically advanced countries around the world; 12 of them financially support the mentioned programs. The leading scientific institutes and foundations as well as best public and private universities in the field of natural sciences are involved in the advanced research and program implementation. For example, the National Quantum Initiative (2018) is being implemented in the USA, which is aimed at maintaining the technological leadership of the USA in the field of quantum technologies in the medium and long term. A series of more than 80 dual-use R&D projects has been deployed under the management of the US National Security Agency, the NSA, the US Agency for Intelligence Advanced Research Projects Activity in the Field of Military Intelligence (IARPA), the Defense Advanced Research Projects Agency of the US Department (DARPA), the US National Science Foundation (NSF), the US Department of Energy, etc. At the same time, the US budget for the development of quantum technologies in 2021 exceeded \$2.5 billion. Other countries are actively participating in international programs for the development of quantum technologies. At the same time, all national quantum programs have been identified by governments as highly important for the state national security and economic competitiveness.

In most national quantum programs, the four following directions of exploratory research are identified:

- quantum computers and computing;
- quantum communications (near-term perspective);
- quantum cryptanalysis (near-term perspective);
- quantum and post-quantum cryptography.

2 Introduction

Therefore, the following key technologies have been identified in a quantum cryptography:

- quantum key distribution (QKD) and quantum encryption in fiber-optic communication channels and in open space;
- quantum hashing and quantum digital signature;
- quantum super-dense encoding of information using “entangled” and “hyper-entangled” particles (one quantum bit (qubit) can carry up to two ordinary bits), which increases the bandwidth of the quantum communication channel;
- encoding in quantum information transmission systems, etc.

In quantum cryptanalysis, Shor’s algorithm provides exponential acceleration of solving factorization problems, discrete logarithm problem (DLP), and elliptic curve discrete logarithm problems (ECDLP), widely used in cryptographic applications in cyberspace. Therefore, the well-known protocols TLS, SSH, IPsec, etc., rely on Diffie–Hellman key agreements (which depend on the strength of DLP or ECDLP), digital signatures (DSA, ECDSA, or RSA-PSS) or public key encryption (El Gamal or RSA-OAEP). Shor’s quantum algorithm can potentially crack most protocols and schemes of asymmetric encryption (public key cryptography). In general, all known quantum algorithms can be divided into two groups: providing exponential gain (for example, Shor’s algorithm) and providing quadratic gain (for example, Grover’s algorithm).

Particular attention is paid to the Shor quantum algorithm and other polynomial algorithms capable of solving cryptanalysis problems with the required reliability and complexity in polynomial time. Grover’s quantum search algorithm is also quite interesting; it speeds up algorithms for solving some NP-class problems, for which a better algorithm than a direct search is unknown: for example, to speed up the key search for crypto systems such as the well-known DES algorithm. The quantum Fourier transformation is also the subject of high interest, which solves the problems of calculating the discrete logarithm and factorization and crack many cryptosystems using a quantum computer, for example, RSA. According to the reports of the American National Institute of Standards (NIST), the AES, SHA-2, SHA-3, RSA, ECDSA, ECDH, and DSA crypto algorithms used in the USA and NATO countries are subject to quantum threat. The quantum computers compute at completely different speeds than modern fifth generation supercomputers, which makes the reading of encrypted texts a very real threat.

For quantum computing, all cryptographies are divided into quantum-secure and quantum-insecure. Algorithms and cryptosystems of symmetric encryption (including AES or GOST R 34.12-2015) are attributed to quantum-secure but with at least twice the key length increased (it is difficult to say exactly what length will be sufficient). Algorithms and cryptosystems of asymmetric encryption based on the complexity of integer factorization (for example, RSA) or discrete logarithm (for example, El Gamal or elliptic curves) should be classified as quantum insecure. Quantum computers also pose a real threat to the security of most well-known blockchain platforms, which universally use asymmetric cryptographic algorithms to create a public-private key pair and an address that is obtained using hashing operations and a checksum of the public key. As a result, in a number of countries, like the USA and the European Union, it is already planned to switch to the use of stable quantum cryptography. Therefore, the mentioned NIST is in the process of developing the quantum cryptography standards, and the NSA recommends its suppliers to implement SHA-384 instead of SHA-256.

This monograph contains the best practice of solving problems of quantum cryptanalysis to improve cybersecurity and resilience of the digital economy. The book discusses the well-known and author's software implementations of promising quantum Shor algorithms, Grover, Simon, and others. Shor's algorithm provides exponential acceleration of solving factorization problems, discrete logarithm (DLP), and elliptic curve discrete logarithm problems (ECDLP). The mentioned tasks are widely used in TLS, SSH, or IPsec cryptographic applications of Internet/Intranet and IIoT/IoT networks, communication protocols based on Diffie-Hellman key agreements (depend on the strength of DLP or ECDLP), digital signature algorithms (DSA, ECDSA, and RSA-PSS), public key encryption algorithms (El Gamal and RSA-OAEP), etc. In other words, Shor's quantum algorithm is potentially capable of violating these algorithms, and with them, all the mechanisms of public key cryptography deployed in cyberspace.

The book will be useful for chief information officers (CIO) and chief information security officers (CISO), certified information systems auditors (CISA), managers of the top echelon of companies responsible for ensuring security and cyber resilience of business, as well as teachers and students of MBA, CIO, and CSO programs, students, and postgraduates of relevant specialties.

The book contains four chapters, dedicated to the following topics:

- The relevance of practical issues of quantum cryptanalysis aimed to improve cybersecurity and resilience of the digital economy.

4 *Introduction*

The analysis of the well-known national quantum programs of technologically developed countries of the world is conducted. The main prerequisites and scientific foundations for solving practical problems of quantum cryptanalysis based on the algorithms of Shor, Grover, Simon, etc., are revealed. The basic requirements for promising quantum cryptanalysis software packages are determined.

- Possible implementations of the Shor factorization algorithm on a quantum circuit to develop requirements for quantum cryptanalysis algorithms. The applicability of Fourier transforms to determine the periodic structure of finite-length sequences is analyzed. The classical factorization problem has been reformulated to solve it on modern quantum computers with 16 or more qubits. A modification of the basic Shor factorization algorithm is proposed for the possibility of its execution on a quantum computer. The complexity of the quantum factorization algorithm is estimated to determine the necessary and sufficient computing resources. An example of a possible implementation of the Shor factorization algorithm on a quantum circuit is presented. The estimation of the limiting capabilities of the quantum factorization algorithm for solving quantum cryptanalysis problems is carried out.
- Quantum cryptanalysis algorithms based on factorization and discrete logarithm problems in various algebraic structures. The stability of the modern cryptographic algorithms has been evaluated to clarify the tasks of cryptanalysis of asymmetric encryption schemes (RSA or El Gamal) and digital signature (DSA, ECDSA, or RSA-PSS) based on factorization and discrete logarithm problems (DLP and ECDLP) in various algebraic structures in a quantum computing model. The necessary and sufficient conditions have been determined for the effective solution of cryptanalysis problems of asymmetric encryption schemes (RSA or El Gamal) and digital signature (DSA, ECDSA, or RSA-PSS) in a quantum computing model. A basic quantum Shor factorization algorithm is proposed. A quantum Grover search algorithm has been developed. A quantum algorithm for symmetric encryption key recovery based on the message text and ciphertext has been developed. A quantum algorithm for cryptanalysis of the RSA asymmetric encryption system has been developed. A quantum algorithm for cryptanalysis of the El Gamal system has been developed.
- Examples of the developed promising software package for quantum cryptanalysis of asymmetric encryption and digital signature

schemes. The well-known quantum emulators Quantum Development Kit, Quantum Computing Playground, jQuantum, QuEST, Quantum Programming Studio, Q-Kit, etc., are considered. The main advantages and limitations of the mentioned quantum emulators are shown. A practical implementation of the Shor algorithm based on the IBM Q platform and the specialized programming language Qiskit is proposed. This made it possible to test a number of possible combined schemes based on the promising IBM Q System One quantum computers with a 20-qubit system, as well as a number of simulations on classic fifth generation supercomputers. The results obtained indicate the feasibility of practical use of the modified Shor algorithm to improve the cryptographic protection of digital economy applications.

Alexei Petrenko

Associate Professor Innopolis University, Ph.D.

May 2022

References

- Abdullin D.I. Shore factorization algorithm based on hardware IP cores in FPGA architecture/FPGA/ Proceedings of the conference. XXIV Tupolev Readings (School of Young Scientists), Kazan, November 07–08, 2019, Kazan National Research Technical University named after A.N. Tupolev-KAI (KNITU-KAI), pp. 573–575.
- Shor's algorithm, its implementation in Haskell and the results of some experiments [Electronic resource]/Programmer's Notes. Access mode: <http://eax.me/shors-algorithm> 14.05.2021 p.
- Andreev, D.Yu., Korzh O.V., Korobkov S.V., Chernyavsky A.Yu. A parallel algorithm for modeling Grover's ideal quantum algorithm. Parallel Computing Technologies (PaVT'2013): Proceedings of the International Scientific Conference (Chelyabinsk, April 1–5, 2013). SUSU Publishing Center, Chelyabinsk, 2013. pp. 38–48.
- Aho A., Hopcroft J., Ulman J. Construction and analysis of computational algorithms: Trans. from English. A.O. Slisenko. / Edited by Yu.V. Matiyasevich, M.: Mir, 1979. 536s.
- Biryukov D.N. Smart cybersecurity/ Biryukov D.N., Petrenko A.S., Petrenko S.A. / Journal: Information Protection. Inside. 2019. No. 4 (88). c. 14–24.
- Biryukov D.N., Lomako A.G., Rostovtsev Yu.G. The appearance of anticipating systems for preventing the risks of implementing cyber threats // Proceedings of SPIIRAN. 2015. No. 2(39). P.5–25.
- Biryukov D.N., Lomako A.G., Rostovtsev Yu.G. An approach to building information security systems capable of synthesizing scenarios of proactive behavior in an information conflict // Information Protection. INSIDE. 2014. No. 6. C. 42–50.
- Bogdanov A.Yu. Quantum algorithms and their impact on the security of modern classical cryptographic systems. / A.Yu. Bogdanov, I.S. Kizhvatov // RGGU. - 2005. 18 p.
- Valiev K.A. Quantum computers and quantum computations / K.A. Valiev. M.: Institute of Physics and Technology, 2005. 387c.
- Vasilenko O. N. Number-theoretic algorithms in cryptography / O. N. Vasilenko. - M.: ICNMO, 2003. 328 p.
- Wentzel, E.S. Probability theory and its engineering applications / E.S. Wentzel, L.A. Ovcharov. M.: Nauka, 1988 480 p.
- Vinberg E. B. Fermat's small theorem and its generalizations // Matem. prosv. M., 2008. vol. 12. pp. 43–53. URL: <http://mi.mathnet.ru/mp238>.
- Wentzel E. S. "Operations research: tasks, principles, methodology"-2nd ed. Moscow: Nauka, 1988. 208 p.
- Gabidulin E. M., Pilipchuk N. I. Lectures on information theory: textbook. Moscow: MIPT, 2007. - 214 p—ISBN 5-7417-0197-3.
- Gulyaeva T. A. Fundamentals of information theory and cryptography. Novosibirsk: NSTU Publishing House, 2010. 88 p. ISBN 978-5-7782-1425-5.
- GOST R 51583-2014 "Information security". The procedure for creating automated systems in a protected version. General provisions". M.: Standartinform. 2014.
- Grishanov S.A. Hardware implementation of a typical module for performing the Shore algorithm in FPGA class FPGA architecture // materials of the scientific and practical seminar. Issue 1. Moscow: IPM named after M.V.Keldysh, 2014. p. 12–16.
- Denisenko D.V., Marshalko G.B., Nikitenkova M.V., Rudskoy V.I., Shishkin V.A. Evaluation of the complexity of implementing the Grover algorithm for sorting the keys of block encryption algorithms GOST R 34.12-2015, Journal of Experimental and Theoretical Physics, RAS, P.L. Institute of Physical Problems. Kapitzy RAS (Moscow), 2019, volume 155, issue 4, pp. 645–653, 2019.
- The Doctrine of Information Security of the Russian Federation [Electronic resource]: approved. By the Decree of the President of the Russian Federation of 5 Dec. 2016 No. 646. URL: <https://rg.ru/2016/12/06/doktrina-infobezobasnost-site-dok.html> (accessed: 05/15/2021)
- Dupliy S.A., Shapoval I.I. Topological methods in quantum computing/ Bulletin of KhNU (ser. "Nuclei, Particles, fields"), 2007, Vol. 781. No. 3(35). c. 3–30. Journal of Kharkov National University, ser. "Nuclei, particles and fields". 2007. V. 781. N 3(35). p. 3–30.
- Dushkin R. Quantum computing and functional programming / R. V. Dushkin. M.: DMK Press, 2015. 232 p.
- Eremeev M.A., Moldovyan N.A., Moldovyan A.A. From primitives to synthesis of algorithms, Publishing House BHV-Petersburg, 2004, 448 p. 2004.

Ishmukhametov Sh.T. Methods of factorization of natural numbers.: textbook / Sh.T. Ishmukhametov. Kazan: Kazan University. 2011. 192 p.

Kalinin, V.N. Theoretical foundations of system research / V.N. Kalinin // St. Petersburg: Mozhaisky VKA. 2013. 278 p.

Kitaev A. , Shen A. , Vyaly M. Classical and quantum computing / M.: ICNMO, Publishing House of Chero, 1999. 192 p.

Clark, E.M. Verification of program models: Model Checking / E.M. Clark , O. Gramberg , D. Peled. Translated from English / Edited by R. Smelyansky . M.: ICNMO, 2002—416 p.

Kolmogorov, A.N. Information theory and theory of algorithms. USSR Academy of Sciences, Moscow: Nauka, 1987.

Kotelnikov V. A. The fate that engulfed the century. In 2 t. / comp. N. V. Kotelnikova. Moscow: Fizmatlit, 2011. 312 p.

Korolkov A.V. On some applied aspects of quantum cryptography in the context of the development of quantum computing and the emergence of quantum computers. / A.V. Korolkov // Issues of cybersecurity No. 1(9) - 2015. M.: Journal "Issues of cybersecurity", 2015. pp. 6–13.

Korzh O.B. , Andreev D.Yu. , Korzh A.A. , Korobkov V.A. , Chernyavsky A.Yu. , Modeling of an ideal quantum computer on a Lomonosov supercomputer, Journal "Computational Methods and Programming", Ed. Research Computing Center of Moscow State University named after M.V. Lomonosov , 2013, No. 14, issue 2, pp. 24–34

Korzh O.B. , Chernyavsky A.Yu. , Korzh A.A. , Simulation of the quantum Fourier transform with noise on the Lomonosov supercomputer, CollectionScientific service on the Internet: all facets of parallelism: Proceedings of the International Supercomputer Conference (September 23–28, 2013, Novorossiysk), Ed. of the Moscow State University . Lomonosova, Moscow, pp. 188–193.

Korn, G. , Handbook of Mathematics for Researchers and Engineers. Definitions, theorems, formulas. / G. Korn , T. Korn // Ed. 4 / Under the general ed. Aramanovich I.G. M. : Nauka, 1978– 832 p.

Klyucharev P.G. Abstract of the dissertation for the Candidate of Technical Sciences. Algorithmic and software for modeling a quantum computer. Moscow State Technical University named after N.E. Bauman , 2009, 18 p.

Knut D. E. The Art of Programming, Volume 2. Calculated methods, 3rd ed. - Williams, 2001— 832 p— ISBN 5-8459-0081-6.408

Knut D. E. The Art of Programming, Volume 2. Sorting and searching., 3rd ed. Williams, 2001. 800 p.

Cryptographic protection of information. The processes of forming and verifying an electronic digital signature [text]: GOST R 34.10-2001. Instead of GOST R 34.10-94, introduction. 01.07.2002. Moscow: IPK Publishing House of Standards, 2001. 16 p— (State Standard of the Russian Federation). — URL: <http://protect.gost.ru/document.aspx?control=7&id=131131>.

Crandall R. , Pomerance K. Prime numbers: Cryptographic and computational aspects / ed. N. Chubarikova ; trans. a. V. Beguntsa [et al.]. M.: URSS: Book house "LIBROCOM", 2011— 664 p.

Manin Yu. I. Computable and non-computable. Moscow: Sovetskoe radio, 1980. 128 p.

Markov, A.S. The experience of identifying vulnerabilities in foreign software products / A.S. Markov , V.L. Cirlov // Questions of cybersecurity. 2013. No. 1(1) pp. 42–48.

Matveev E.A. Dissertation for the Candidate of Physical and Mathematical Sciences. Application of quantum mechanical effects in information security systems. Penza, NTP Cryptosoft, 2019 157 p.

Mesarovich, M. Theory of hierarchical multilevel systems / M. Mesarovich , D. Mako , I. Takahara . M.: Mir, 1973. p. 344.

Moldovyan A.A. , Moldovyan N.A. New forms of the hidden discrete logarithm problem. Proceedings of SPIIRAN 2019. Volume 18 No. 2. Pp. 504–529.

Moldovyan N.A. , *Introduction to Public Key Cryptosystems* / N.A. Moldovyan , Moldovyan A.A. /, Publishing House of BHV-Petersburg, 2005, 286 p. 2005.

Moldovyan N.A. , Workshop on cryptosystems with a public key, Publishing House BHV-Petersburg, 2005, 298 p. 2007.

Nikolenko S.I. New constructions of cryptographic primitives based on semigroups, groups and linear algebra. Dissertation for the Candidate of Physical and Mathematical Sciences. St. Petersburg, Institution of the Russian Academy of Sciences St. Petersburg Department of the Mathematical Institute named after V.A. Steklova RAS, 2008 120 p.

Nielsen M. , Chang I. Quantum computing and quantum information. Translated from English: Mir, 2006 824 p., ill.

Fundamentals of cryptography. Textbook / A. P. Alferov [et al.]. - M.: Helios ARV, 2001. 480 p— ISBN 5-85438-137-0.

Petrenko, A.S. , Romanchenko A.M. A promising method of cryptanalysis based on the Shor algorithm// Information security. Inside No. 2 2020. St. Petersburg: Publishing house Athena, 2020. pp. 17–23.

Petrenko A.S. On the implementation of the partially homomorphic El-Gamal cryptosystem. In the collection: The 2019 Symposium on Cybersecurity of the Digital Economy (CDE19). The third International Scientific and Technical Conference. Saint Petersburg, 2019. pp. 262–265.

Petrenko A.S. On the implementation of the partially homomorphic RSA cryptosystem. In the collection: The 2019 Symposium on Cybersecurity of the Digital Economy (CDE19). The third International Scientific and Technical Conference. Saint Petersburg, 2019. pp. 266–268.

Petrenko A.S. On the implementation of the partially homomorphic Paie cryptosystem. In the collection: The 2019 Symposium on Cybersecurity of the Digital Economy (CDE19). The third International Scientific and Technical Conference. Saint Petersburg, 2019. pp. 266–268.

Petrenko A.S. On the implementation of a completely homomorphic Gentry-Halevi-Smart cryptosystem. In the collection: The 2019 Symposium on Cybersecurity of the Digital Economy (CDE19). The third International Scientific and Technical Conference. Saint Petersburg, 2019. pp. 272–275.

Petrenko A.S. On the implementation of a fully homomorphic cryptosystem based on matrix polynomials. In the collection: The 2019 Symposium on Cybersecurity of the Digital Economy (CDE19). The third International Scientific and Technical Conference. Saint Petersburg, 2019. Pp. 276–279.

Petrenko S.A. , Stupin D.D. National early warning system about a computer attack: a scientific monograph / edited by S. F. Boeva. Publishing House "Athena", 2018– 440 p.

Petrenko S. Cyber Resilient Platform for Internet of Things (IIoT/IoT)ed Systems: Survey of Architecture Patterns. Voprosy Kiberbezopasnosti [Cybersecurity issue]. 2021. No 2 (42). P. 81–91. DOI: 10.21681/2311-3456-2021-2-81-91

Petrenko S. Self-Healing Cloud Computing. Voprosy Kiberbezopasnosti [Cybersecurity issue]. 2021. No 1 (41). P. 80–89. DOI: 10.21681/2311-3456-2021-1-80-89.

Pravilshchikov P.A. Quantum parallelism and the solution of equations in control problems based on a new model of computing / P.A. Pravoshchikov. M.: V.A. Institute of Management Problems. Trapeznikova, 2014. 179 p.

Order No. 31 of the FSTEC of Russia dated 14.03.2014 "On Approval of requirements for ensuring the protection of Information in Automated Control Systems for Production and Technological Processes at Critical Facilities, Potentially Dangerous Facilities, as well as Facilities that pose an increased danger to Human Life and Health and to the Environment". Access mode: <http://fstec.ru>.

Order No. 235 of the FSTEC of Russia dated December 21, 2017 "On Approval of Requirements for the Creation of Security Systems for Significant Objects of the Critical Information Structure of the Russian Federation and Ensuring their Functioning". Access mode: <http://fstec.ru>.

Order No. 239 of the FSTEC of Russia dated December 25, 2017 "On Approval of Requirements for Ensuring the Security of Significant Objects of the Critical Information Infrastructure of the Russian Federation". Access mode: <http://fstec.ru>.

Preskill J. Quantum information and quantum computations / D. Preskill . M.: Izhevsk, 2008. 464 p.

Rostovtsev, Yu.G. Fundamentals of building automated systems for collecting and processing information. - Ministry of Defense of the Russian Federation, 1992– 717 p.

Guidance document. Computer equipment. Protection against unauthorized access to information Indicators of security against unauthorized access to information. Approved by the decision of the Chairman of the State Technical Commission under the President of the Russian Federation dated March 30, 1992. Access mode: <http://fstec.ru>.

Guidance document. Protection against unauthorized access to information. Part 1. Information security software. Classification according to the level of control of the absence of undeclared opportunities. Approved by the decision of the Chairman of the State Technical Commission under the President of the Russian Federation dated March 30, 1992 Access mode: <http://fstec.ru>.

Ruchkin V.N. Natural parallelism of quantum computers and neural calculators / V.N. Ruchkin , V.A. Romanchuk , V.A. Fulin. Ryazan.: Ryazan State University named after S.A. Yesenina, 2013 387 p.

Ryzhikov, Yu.I. Computational methods / Yu.I. Ryzhikov // Ucheb. pos. -SPb: BHV-Petersburg. - 2007. 397 p.

Saati, Vol. . Decision-making. Method of hierarchy analysis // M: Radio and Communications. 1993. 278 p.

Sachkov V.N. , V.A. Kotelnikov and encrypted communication. Conferences and Symposiums, vol. 176, No. 7, UFN 2006, pp. 775–777

Information processing systems. Cryptographic protection. Algorithm of cryptographic transformation [text]: GOST 28147-89. Introduction. 01.07.90. Moscow: Publishing House of Standards, 1989. 28 p— (State standard of the USSR). — URL: <http://protect.gost.ru/document.aspx?control=7&id=139177>.

Dictionary of Cryptographic Terms. Edited by B. A. Pogorelova and V. N. Sachkova. Moscow State University named after M.V. Lomonosov Academy of Cryptography of the Russian Federation. Moscow ICNMO Publishing House. 2006— 50 p.

The National Security Strategy of the Russian Federation (approved by Decree The President of the Russian Federation on December 31, 2015 N 683) // SPS "ConsultantPlus".

Tokareva N.N. On the History of Cryptography in Russia, Historical Essays on Discrete Mathematics and its Applications, No. 4 (18), S.L. Mathematical Institute. Soboleva SB RAS Novosibirsk, 2012, pp. 82–107.

Thomas H. Algorithms. Construction and analysis [Text]/ X. Thomas - M. : Williams, 2016. 1328 p.

Requirements for ensuring the security of significant objects of the critical information infrastructure of the Russian Federation, approved by the Order of the FSTEC of Russia No. 239 dated December 25, 2017 Access mode: <http://fstec.ru>.

Federal Law No. 187-FZ of June 26, 2017 "On the Security of the Critical Information Structure of the Russian Federation". Access mode: <https://rg.ru>.

Federal Law No. 149-FZ of 27.07.2006 "On Information, Information Technologies and Information Protection". Access mode: <https://rg.ru>.

Kholevo A. S. Mathematical foundations of quantum computer science M.: MIAN, 2018. 118 p— (Lecture. REC courses, ISSN 2226-8782; Issue 30). ISBN 978-5-98419-080-7

Kholevo A. S. Quantum systems, channels, information. Electronic edition. Moscow: ICNMO, 2014. 327 p. ISBN 978-5-4439-2092-4

Kholevo A. S. . Introduction to quantum information theory. MCNMO, 2002 128 p.

Cheremushkin A. V. Cryptographic protocols: basic properties and vulnerabilities // Applied Discrete Mathematics. 2009. Nov. issue 2. pp. 115–150. — URL: <https://cyberleninka.ru/article/n/kriptograficheskieprotokoly-osnovnye-svoystva-i-uyazvimosti.pdf>.

Shalagin S.V. Implementation of the Shore algorithm based on IP cores in PLIS/FPGA //C architecture. V. Shalagin , S.A. Grishanov // Fundamental and applied problems of mathematics and computer science. Materials of the XII International Conference. Makhachkala: publishing house of DSU, 2017. pp. 202–205.

Shannon K. Works on information theory and cybernetics / edited by R. L. Dobrushina , O. B. Lupanova . M.: Publishing House of Foreign Literature, 1963— 830 p.

Schneier B. Applied cryptography: protocols, algorithms, source code in the C language. Williams Publishing House, 2016— 816 p.

Ebbinhaus G.D. Turing machines and recursive functions/ G.D. Ebbinhaus . M.: Mir, 1972–264 p.

Shor P. Algorithms for quantum computation: discrete logarithms and factoring [Text] /Shor P.// Foundations of Computer Science.— 1994.— No. 10. —134p.

Aaronson, Scott (2013). Quantum Computing Since Democritus. Cambridge: Cambridge University Press.

Aaronson S. , Arkhipov A. , The Computational Complexity of Linear Optics, Proceedings of the forty-third annual ACM symposium on Theory of computing, 333–342 (2011)

Arute F. , Arya K. , Martinis John M. , Quantum supremacy using a programmable superconducting processor, Nature, 574, 505–510 (2019)

Bell, Philip (2018). *Beyond Weird: Why Everything You Knew About Quantum Physics Is Different*. Chicago: University of Chicago Press.

Benioff P , The computer as a physical system: A microscopic quantum mechanical Hamiltonian model of computers as represented by Turing machines J. Statist. Phys. 22, 563–591 (1980)

Bernhardt, Chris (2019). *Quantum Computing for Everyone*. Cambridge, MA: MIT Press.

Boulund A. , Fefferman B. , Nirkhe Ch. , Quantum Supremacy and the Complexity of Random Circuit Sampling, arxiv.org/abs/1803.04402 (2018)

Carroll, Sean (2019). *Something Deeply Hidden*. United States: Dutton.

Deutsch D. , Quantum theory, the Church-Turing principle and the universal quantum computer, *Proceedings of the Royal Society A*. 400 (1818), 97–117 (1985)

Deutsch D. , Jozsa R. , Rapid solution of problems by quantum computation, *Proceedings of the Royal Society of London A*, 439, (1907), 553–558 (1992)

Diffie D , Hellman M . New directions in cryptography, *IEEE Transactions on Information Theory*, v. 22, Issue 6 (1976)

Feynman R , Simulating physics with computers, *Internat. J. Theoret. Phys.* 21, 467–488 (1982)

Grover L.K. , A fast quantum mechanical algorithm for database search, In *Proceedings of the twenty-eighth, annual ACM symposium on Theory of computing*, 212–219, ACM (1996)

Huang C , Zhang F. , Newman M / , Classical Simulation of Quantum Supremacy Circuits, arxiv.org/abs/2005.06787 (2020) 19.

A. Zlokapa , S. Boixo , D. Lidar , Boundaries of quantum supremacy via random circuit sampling, arxiv.org/abs/2005.02464 (2020)

Johnston, Eric R. , Nic Harrigan , and Mercedes Gimeno-Segovia (2019). *Programming Quantum Computers: Essential Algorithms and Code Elements*. Sebastopol, CA: O'Reilly.

Kumar, Manjit (2011). *Quantum: Einstein, Bohr, and the Great Debate about the Nature of Reality*. New Delhi: Hachette India.

Orzel, Chad (2009). *How to Teach [Quantum] Physics to Your Dog*. New York: Scribner.

Orzel, Chad (2018). *Breakfast with Einstein: The Exotic Physics of Everyday Objects*. Dallas, TX: BenBella Books, Inc.

Pednault E , Gunnel J. , On “Quantum Supremacy”, ibm.com/blogs/research/2019/10/on-quantum-supremacy (2019)

Rhodes, Richard (2012). *Hedy's Folly: The Life and Breakthrough Inventions of Hedy Lamarr, the Most Beautiful Woman in the World*. New York: Doubleday.

Shor, P.W. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer, *SIAM J. Computing* 26, 1484–1509 (1997)

Simon D.R. , On the power of quantum computation, *SFCS '94: Proceedings of the 35th Annual Symposium on Foundations of Computer Science*, 116–123 (1994)

Zhou Y , Stoudenmire E.M. , Waintal X. , What limits the simulation of quantum computers?, arxiv.org/abs/2002.07730 (2021)

Magomedov S. , Gusev A. , Ilin D. , Nikulchev E. Used the Time of User Reactions to Improve Security and Control Access To Web Services // *Applied Science*. 2021. Vol. 11, No. 5. P. 2561.

Magomedov S , Ilin D , Nikulchev E . Resource Analysis of the Log Files Storage Based on Simulation Models in a Virtual Environment. // *Applied Science*. 2021. Vol. 11, No. 11. P. 4718.

Magomedov S. , Lebedev, A. Protected network architecture for ensuring consistency of 2 medical data through validation of user behavior and DICOM 3 archive integrity // *Applied Science*. 2021, Vol. 11. No. 5. P. 2072.

Magomedov S. , Ilin D. , Silaeva A. , Nikulchev E. Dataset of User Reactions When Filling Out Web Questionnaires // *Data*. 2020. Vol. 5, No. 4. P. 108, 1–107.

Nikulchev E. , Ilin D. , Silaeva A. , Kolyasnikov P. , Belov V. , Runtov A. , Pushkin P. , Laptev N. , Alexeenko A. , Magomedov S. , Kosenkov A. , Zakharov, I. ; Ismatullina, V. ; Malykh, S. Digital Psychological Platform for Mass Web-Surveys // *Data*. 2020. V. 5, No. 4. P. 95, 1–16.

Moody Dustin, NIST PQC team , NIST Status Update on the 3rd Round. Cryptographic Technology Group. Computer Security Division. 2021, <https://csrc.nist.gov/CSRC/media/Presentations/status-update-on-the3rd-round/images-media/session-1-moody-nist-round-3-update.pdf>

Belenko V. , Krundyshev V. , Kalinin M. Synthetic datasets generation for intrusion detection in VANET // *Proceedings of the 11th International Conference on Security of Information and Networks*. 2018.

Myasnikov A.V. Application of machine learning technologies to optimize the penetration testing process // Problems of information security. Computer systems. 2019. No. 2. p.9

Dakhnovich A.D. , Moskvina D.A. , Zegzhda D.P. Analysis of threats to information security in digital production networks // Problems of information security. Computer systems. 2017. No. 4. pp.41–46.

Zegzhda P.D. , Rudina E.A. Fundamentals of information security. Textbook Publishing House of the Polytechnic University, St. Petersburg, 2008, 224 p.

Poltavtseva M.A. , Zegzhda D.P. , Suprun A.F. Database security. Textbook Publishing House of the Polytechnic University, St. Petersburg, 2008, 224 p.

Alexandrova E.B. , Shkorkina. E.N. , Oblogina A.Yu . Authentication of control devices in the Internet of Things network with the architecture of boundary computing // Problems of information security. Computer systems. – 2021. No. 2.

Aleksandrova E. B. , Shkorkina E. N. Using Undeniable Signature on Elliptic Curves to Verify Servers in Outsourced Computations // Automatic Control and Computer Sciences. 2018. V. 52. No. 8. P. 1160–1163.

Zegzhda D.P. , Alexandrova E.B. , Kalinin M.O. , etc. Cybersecurity of the digital industry. Theory and practice of functional resistance to cyber attacks. Publishing house: Hotline - Telecom, 2020, 560 pages.

Vorobyev E.G. , Khomonenko A.D. Mathematical model of binary-decimal conversion and algorithms of information compression based on linear functions. // "Information protection. Inside." 2019. No. 5. pp.15–21.

Vorobyov E. G. , Khomonenko A. D. Models of vector representation of multi-bit binary data based on pseudoregular numbers // Control, communication and security systems. 2019. No. 2. pp. 291–303.

Vorobyov E.G. Mathematical models of management of the system for ensuring the availability of information and assessing the quality of its functioning // High-tech technologies in Earth space research (H&ES Research). - 2019. No. 2. pp. 51–64.

Vorobyov E.G. Compression methods based on reversible calculations. // Izvestiya SPbGETU "LETI". 2015. No. 5. pp. 15–21.

Vorobyov E.G. Compression of binary codes based on traditional methods and the use of pseudoregular numbers. // Izvestiya SPbGETU "LETI". 2015. No. 5. p. 23–28.

Vorobyov E.G. The concept of ensuring the continuity of the functioning of distributed information and telecommunication systems based on quantum methods of information representation. // Izvestiya SPbGETU "LETI". 2015. No. 4. pp. 17–21.

Vorobyov E.G. Using the fractal structure of fields formed by codes with different bases to solve the problem of creating a single information space. // Izvestiya SPbGETU "LETI". 2015. No. 3. pp. 11–16.

Vorobyov E.G. Complex numbers and optimization of information storage facilities in global information systems. // Izvestiya SPbGETU "LETI". 2015. No. 2. pp. 22–25.

Vorobyov E.G. Calculation of the effectiveness of information attacks by an external intruder on informatization objects with a distributed infrastructure. // Izvestiya SPbGETU "LETI". 2015. No. 1. p.23–28.

Vorobyov E.G. Controlled polarization of electromagnetic waves as a means of increasing the secrecy of information transmission. // Izvestiya SPbGETU "LETI". 2014. No. 9. pp. 44–49.

Vorobyov E.G. Tsekhanovsky V.V. Pseudoregular numbers in binary fields. // Izvestiya SPbGETU "LETI". 2014. No. 2. pp. 18–23.

D. A. Nabokov , "Post-quantum electronic voting based on grids with the participation of several candidates", PDM. Appendix, 2021, No. 14, 95–100

M. A. Kudinov , E. O. Kiktenko , A. K. Fedorov , Security analysis of the W-OTS+ signature scheme: Updating security bounds Mathematical Issues of Cryptography, 12:2 (2021), 129–145

Grebnev S.V. , Klyucharev P.G. , Koreneva A.M. , Koshelev D.I. , Taraskin O.G. , Tulebaev A.I. A post-quantum cryptographic protocol for generating a common key based on isogeny of supersingular elliptic curves. Proceedings of the XI International Scientific and Technical Conference "Secure Information Technologies". Moscow State Technical University named after N.E. Bauman, 2021.

S.E. Yunakovskiy , M. Kot , N.O. Pozhar , D. Nabokov , M.A. Kudinov , A. Guglya , E.O. Kiktenko , E. Kolycheva , A. Borisov , A.K. Fedorov . Towards security recommendations for public-key infrastructures for production environments in the post-quantum era. EPJ Quantum

Technology (2021) arXiv: 2105.01324

E.O. Kiktenko , M.A. Kudinov , and A.K. Fedorov . Detecting bruteforce attacks on cryptocurrency wallets, Lecture Notes in Business Information Processing. 373, 232–242 (2019); arXiv:1904.06943

E.O. Kiktenko , M.A. Kudinov , A.A. Bulychev , and A.K. Fedorov . Proof-of-forgery for hash-based signatures. arXiv:1905.12993

E.O. Kiktenko , A.A. Bulychev , P.A. Karagodin , N.O. Pozhar , M.N. Anufriev , and A.K. Fedorov . SPHINCS+ digital signature scheme with GOST hash functions. arXiv:1904.06525
M.A. Kudinov , E.O. Kiktenko , and A.K. Fedorov . Security analysis of the W-OTS+ signature scheme: Updating security bounds. arXiv:2002.07419S.

S.V. Grebnev . Limonnitsa: making Limonnik-3 post-quantum. Mathematical Questions of Cryptography, 11:2 (2020), 25–42

Grebnev S.V. Post-quantum cryptography: Trends, problems and prospects. Information Security. 2019, 2.

E.O. Kiktenko , N.O. Pozhar , M.N. Anufriev , A.S. Trushechkin , R.R. Yunusov , Y.V. Kurochkin , A.I. Lvovsky , and A.K. Fedorov , Quantumsecured blockchain, Quantum Science and Technology 3, 035004 (2018); arXiv:1705.09258

A.K. Fedorov , E.O. Kiktenko , and A.I. Lvovsky . Quantum computers put blockchain security at risk Nature 563, 465 (2018);

M. A. Kudinov , A. A. Chilikov , E. O. Kiktenko , A. K. Fedorov . Advanced attribute-based protocol based on the modified secret sharing scheme Journal of Computer Virology and Hacking Techniques volume 16, pages 333–341 (2020) arXiv:1912.03009

Grebnev S.V. On the cryptographic properties of the scheme for generating a common "Limonnik-3" key. Information technology security. 2019. 2(27). 6–20

S. V. Grebnev , E. V. Lazareva , P. A. Lebedev , A. Yu. Nesterenko , A. M. Semenov. Integration of domestic protocols for generating a common key into the TLS 1.3 protocol. PDM. Appendix, 2018, 11, 62–65.

S. V. Grebnev . On the feasibility of an ECDLP algorithm Cryptology ePrint Archive, Report 2018/399

D.A. Kronberg , E.O. Kiktenko , A.K. Fedorov , and Y.V. Kurochkin , Analysis of coherent quantum cryptography protocol vulnerability to an active beam-splitting attack Quantum Electronics 47, 163–168 (2017); arXiv:1611.04112

A.S. Trushechkin , E.O. Kiktenko , and A.K. Fedorov , Practical issues in decoy-state quantum key distribution based on the central limit theorem Physical Review A 96, 022316 (2017); arXiv:1702.08531

E.O. Kiktenko , A.S. Trushechkin , C.C.W. Lim , Y.V. Kurochkin , and A.K. Fedorov , Symmetric blind information reconciliation for quantum key distribution Physical Review Applied 8, 044017 (2017); arXiv:1612.03673

A.S. Trushechkin , P.A. Tregubov , E.O. Kiktenko , Y.V. Kurochkin , and A.K. Fedorov , Quantum-key- distribution protocol with pseudorandom bases Physical Review A 97, 012311 (2018); arXiv:1706.00611

A. Farouk , J. Batle M. Elhoseny , M. Naseri , M. Lone , A.K. Fedorov , M. Alkhambashi , S.H. Ahmed , and M. Abdel-Aty , Robust general N user authentication scheme in a centralized quantum communication network via generalized GHZ states Frontiers of Physics 13, 130306 (2018).

A.K. Fedorov , E.O. Kiktenko , and A.S. Trushechkin . Symmetric blind information reconciliation and hash-function-based verification for quantum key distribution Lobachevskii Journal of Mathematics 39, 992 (2018); arXiv:1705.06664

E.O. Kiktenko , A.O. Malyshev , A.A. Bozhedarov , N.O. Pozhar , M.N. Anufriev , and A.K. Fedorov , Error estimation at the information reconciliation stage of quantum key distribution Journal of Russian Laser Research 39, 558 (2018); arXiv:1810.05841

A.K. Fedorov , I. Gerhardt , A. Huang , J. Jogenfors , Y. Kurochkin , A. Lamas-Linares , J.-A. Larsson , G. Leuchs , L. Lydersen , V. Makarov , and J. Skaar . Comment on "Inherent security of phase coding quantum key distribution systems against detector blinding attacks" [Laser Phys. Lett. 15, 095203 (2018)], Laser Physics Letters 16, 019401 (2019); arXiv:1809.03911

E.O. Kiktenko , N.O. Pozhar , A.V. Duplinskiy , A.A. Kanapin , A.S. Sokolov , S.S. Vorobey , A.V. Miller , V.E. Ustimchik , M.N. Anufriev , A.S. Trushechkin , R.R. Yunusov , V.L. Kurochkin , Y.V. Kurochkin , and A.K. Fedorov , Demonstration of a quantum key distribution network in

urban fibre-optic communication lines Quantum Electronics 47, 798–802 (2017); arXiv:1705.07154

A.V. Duplinskiy , E.O. Kiktenko , N.O. Pozhar , M.N. Anufriev , R.P. Ermakov , A.I. Kotov , A.V. Brodskiy , R.R. Yunusov , V.L. Kurochkin , A.K. Fedorov , and Y.V. Kurochkin . Quantum-secured data transmission in urban fibre-optic communication lines Journal of Russian Laser Research 39, 113 (2018); arXiv:1712.09831

V.E. Rodimin , E.O. Kiktenko , V.V. Usova , M.Yu. Ponomarev , T.V. Kazieva , A.V. Miller , A.S. Sokolov , A.A. Kanapin , A.V. Losev , A.S. Trushechkin , M.N. Anufriev , N.O. Pozhar , V.L. Kurochkin , Y.V. Kurochkin , and A.K. Fedorov , Modular quantum key distribution setup for research and development applications, Journal of Russian Laser Research 40, 221 (2019); arXiv:1612.04168

A.K. Fedorov , A.A. Kanapin , V.L. Kurochkin , Y.V. Kurochkin , A.V. Losev , A.V. Miller , I.O. Pashinskiy , V.E. Rodimin , and A.S. Sokolov , Educational potential of quantum cryptography and its experimental modular realization, Proceedings of the Scientific-Practical Conference "Research and Development – 2016", 83 (2018); arXiv:1710.08090

R. Shakhovoy , D. Sych , V. Sharoglazova , A. Udaltsov , A.K. Fedorov , and Y. Kurochkin , Quantum noise extraction from the interference of laser pulses in an optical quantum random number generator, Optics Express 28, 6209 (2020); arXiv:1910.00219

I.S. Kabanov , R.R. Yunusov , Y.V. Kurochkin , and A.K. Fedorov . Practical cryptographic strategies in the post-quantum era, AIP Conference Proceedings 1936, 020021 (2018); arXiv:1703.04285

V. S. Belsky , I. V. Chizhov , A. A. Chichaeva , V. A. Shishkin . Physically unclonable functions in cryptography. International Journal of Open Information Technologies ISSN: 2307-8162 vol. 8, no.10, 2020

A.A. Garage , I.Yu. Gerasimov , M.V. Nikolaev , I.V. Chizhov. On the use of fully homomorphic encryption libraries. International Journal of Open Information Technologies ISSN: 2307-8162 vol. 9, no.3, 2021